CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in turning fragmented enterprise data into structured data models with AI on top, from extraction and matching to evaluation and deployment, working directly with domain experts. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in Python, SQL and machine learning across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end at an early-stage start-up: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) that turns unstructured data from regulatory and technical documents into a knowledge-graph data model, raising precision/recall from 33%/45% to 93%/90% against a ground truth derived from enterprise systems (SAP).
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, SQL, Postgres, Neo4j, MongoDB, LangGraph, LangChain, LiteLLM, GraphRAG, RAG, MCP, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Consulting role: worked with business stakeholders to understand their decision-making workflows, advised where agentic AI could support them and translated requirements into the solution: a LangGraph application routing requests across vector database queries, LLM inference and summarization.
- Developed a Python backend with FastAPI and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.

- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: BOM Emission Factor Matcher | Python | SQL | Postgres | LLM | Streamlit

- Customer-style implementation: Python + SQL data model over synthetic ERP exports (material master, multi-level bills of materials, transactional data from purchase lines) that matches items to public emission factors through exact, rule, embedding and LLM steps, with data quality checks and match accuracy on a labelled set.
- Streamlit view rolling footprints up through the BOM with a summary for executives of the largest supplier levers.

TECHNICAL SKILLSET

- **Data:** Python, SQL, Postgres, MongoDB, Neo4j, SPARQL, Pandas, NumPy, data modeling, Streamlit
- **AI:** LLMs, LangGraph (multi-agent), LangChain, LiteLLM, RAG, GraphRAG, embeddings, MCP, Scikit-Learn, XGBoost, PyTorch, evaluation (RAGAS)
- **Engineering & Cloud:** FastAPI, REST APIs, Docker, Kubernetes, Azure, AWS (CDK, Lambda, SQS, DynamoDB), Git, GitLab CI/CD
- **Ways of working:** stakeholder workflow mapping, AI consulting, Agile, AI-assisted development (Claude Code)
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI